

DSA210 Introduction to Data Science

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Project Title: Train Delay Analysis and Prediction in Lausanne

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Contents

1	Introduction	1
2	Motivation	1
3	Data Sources and Collection	2
3.1	Train Data	2
3.2	Weather Enrichment	2
3.3	Final Dataset	2
4	Data Analysis Pipeline	3
4.1	Data Preparation	3
4.2	Exploratory Data Analysis	3
4.3	Hypothesis Testing	3
4.4	Machine Learning	4
5	Findings	4
5.1	Descriptive Patterns	4
5.2	Hypothesis Test Results	5
6	Modeling Results and Error Analysis	6
6.1	Model Comparison	6
6.2	Overall Error Profile	7
6.3	Where the Model Struggles	7
7	Limitations and Future Work	9

1 Introduction

This project studies whether temporal, operational, and weather factors are associated with train delays in the Lausanne area and whether those factors can support a supervised model for delay prediction. I combine SBB/CFF Istdaten train-arrival records with hourly Open-Meteo weather observations, construct a 90,818-row analysis dataset, run hypothesis tests, and compare six classifiers. The strongest statistical differences are linked to scheduled hour, weekday, season, weekend status, train type, line, station, rain, temperature group, and wind group. XGBoost performs best among the tested models, reaching 0.648 test accuracy, 0.642 delayed-class F1, and 0.710 ROC-AUC. Error analysis shows that the model is most uncertain around specific busy or transition periods: the 19:00 hour has the highest hour-level error rate, rush periods create many false delay alarms, and midday hours miss a larger share of true delays.

2 Motivation

Train delays are small events individually, but they affect daily mobility at a city scale. A few minutes of delay can change transfer plans, crowd platforms, and make a timetable feel less reliable to commuters. Lausanne is a useful case study because it has a mix of local, regional, intercity, and international train services, and because the station area is exposed to seasonal weather changes. The project is also personally relevant because I will start my doctoral studies in Lausanne next year, so understanding the reliability of the local rail network connects directly to the city I am about to live and work in.

The project asks:

Which operational, temporal, and weather features are associated with higher train delay rates in Lausanne, and how well can those features predict whether a specific arrival will be delayed?

The first part of the question is explanatory: it focuses on patterns that can be described and tested statistically. The second part is predictive: it asks whether those same patterns can be used to construct a machine learning model for predicting delays.

3 Data Sources and Collection

3.1 Train Data

The main data source is SBB/CFF Istdaten, the public actual-transport dataset published through OpenTransportData Switzerland. The raw records contain planned and actual or predicted arrival and departure times, operator fields, line information, station names, and train descriptors. I used the official archive index to collect daily CSV files from four 30-day windows: January, April, July, and October 2025. These months were chosen to cover all the seasons while keeping the dataset manageable for analysis.

The raw SBB files are large monthly archives. The collection notebook can download and extract them, but it skips downloading by default once the local daily CSV files already exist. I filtered the raw records to arrivals in the Lausanne area and kept only rows with valid scheduled and predicted arrival times.

3.2 Weather Enrichment

I enriched the train records with Open-Meteo historical hourly weather data for Lausanne. The weather fields used in the final dataset are temperature, precipitation, snowfall, and wind speed. Each train arrival is matched to the weather observation from the same scheduled date and scheduled arrival hour.

3.3 Final Dataset

After filtering, feature engineering, and weather merging, the final dataset has 90,818 arrival-level rows and 21 columns. The target variable is **delayed**, where 1 means the predicted arrival time is later than the scheduled arrival time. Table 1 summarizes the final analysis dataset.

Table 1: Final dataset summary.

Item	Value
Final arrival rows	90,818
Delayed arrivals	43,058 (47.4%)
On-time or early arrivals	47,760 (52.6%)
Matched weather coverage	100.0%
Train/test split	72,654 / 18,164 rows

The final feature set includes scheduled arrival hour, weekday, season, weekend status, station, train type, line name, operator, and the four weather variables. The final merge has complete weather coverage: every train-arrival row has a matched hourly weather record.

4 Data Analysis Pipeline

The analysis is organized as four reproducible notebooks. Each notebook writes its outputs to the repository so later stages can be inspected without rerunning the entire pipeline.

4.1 Data Preparation

The first notebook, `data_collection.ipynb`, performs data collection, filtering, translation, feature engineering, weather collection, and the final merge. The most important variables are:

- `delay_minutes`: predicted arrival minus scheduled arrival, measured in minutes.
- `delayed`: binary target equal to 1 when `delay_minutes > 0`.
- temporal features such as scheduled arrival hour, weekday, month, season, and weekend status.
- operational features such as station, train type, line name, and operator.
- hourly weather features: temperature, precipitation, snowfall, and wind speed.

4.2 Exploratory Data Analysis

The second notebook, `eda.ipynb`, profiles the dataset and visualizes delay patterns by time, service structure, and weather. Since the target is binary, most descriptive plots use the mean of `delayed` as the delay rate. The EDA also compares delay minutes, class balance, hourly weather distributions, and interaction views such as rain effects by hour and line.

4.3 Hypothesis Testing

The third notebook, `hypothesis_testing.ipynb`, turns the strongest EDA patterns into formal tests. Categorical features are tested with chi-square independence tests. Weather variables are aggregated to matched weather hours, and hourly delay rates are compared with one-way ANOVA after binning continuous variables into groups. The significance threshold is $\alpha = 0.05$.

4.4 Machine Learning

The fourth notebook, `ml_modeling.ipynb`, creates an 80/20 train/test split stratified by the joint `season × delayed` key. This preserves both seasonal composition and class balance across the split. The same key is used to create five saved cross-validation folds inside the training set.

To avoid test set leakage, hypothesis tests are repeated on the training data only for feature selection. These tests select ten features: weekend status, line name, precipitation, scheduled arrival hour, weekday, season, station, temperature, train type, and wind speed. Operator and snowfall are not selected. Six models are then compared: a majority-class baseline, logistic regression trained with stochastic gradient descent, k-nearest neighbors, decision tree, random forest, and XGBoost. Since the delayed share is close to 50% (the dataset is close to balanced), hyperparameter tuning uses accuracy as the primary metric.

5 Findings

5.1 Descriptive Patterns

The dataset is close to balanced: 47.4% of arrivals are delayed and 52.6% are on time or early. Delay rates vary visibly across scheduled hour, weekday, train type, line, and station. Figure 1 summarizes three of the clearest descriptive views.

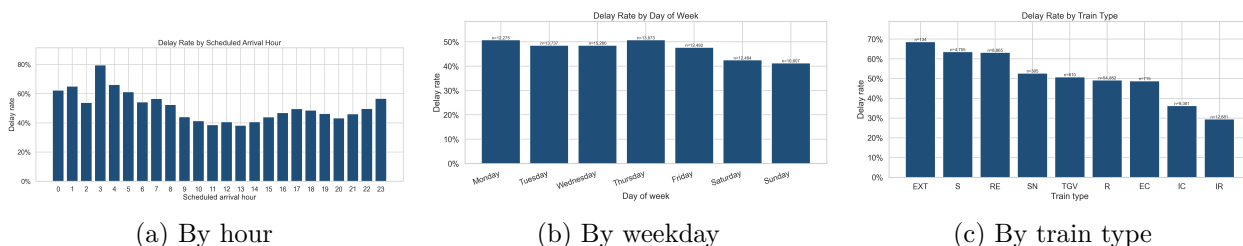
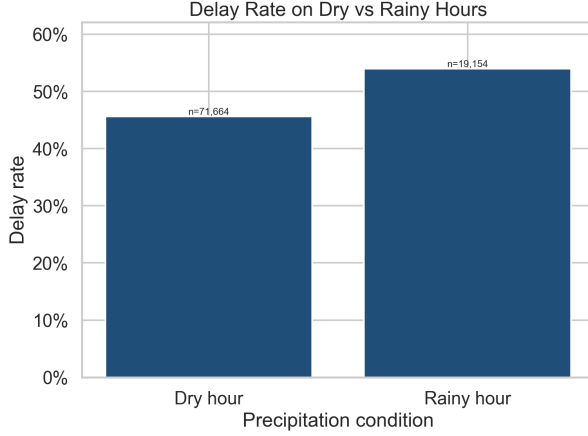
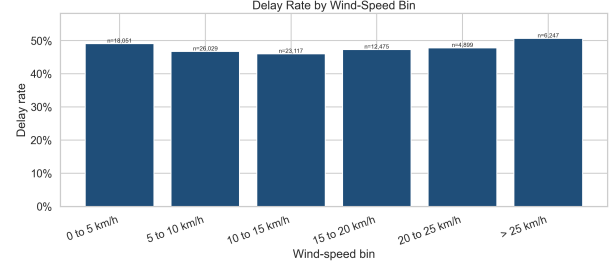


Figure 1: Selected EDA delay-rate patterns.

Weather is also related to delay rates, but its patterns are more subtle than the operational and temporal variables. Rainy hours show a higher delay rate than dry hours in the descriptive analysis. Temperature and wind groups also show differences, while snowfall appears weaker in this sample.



(a) Rainy vs dry hours



(b) Wind speed groups

Figure 2: Weather-related descriptive patterns.

5.2 Hypothesis Test Results

Table 2 gives a compact view of the tests. At the 0.05 level, the project rejects the null hypothesis for scheduled arrival hour, weekday, season, weekend status, train type, line name, temperature group, rainy versus dry hour, wind speed group, and station. The tests do not reject the operator comparison or the snowy versus non-snowy hour comparison.

Table 2: Compact hypothesis-test results.

ID	Test topic	<i>p</i> -value	Decision
H1	scheduled arrival hour	<0.001	Reject
H2	day of week	<0.001	Reject
H3'	season	<0.001	Reject
H4	weekend status	<0.001	Reject
H5	train type	<0.001	Reject
H6	line names	<0.001	Reject
H7	operators	0.201	Do not reject
H8	across temperature groups	<0.001	Reject
H9	between rainy and dry hours	<0.001	Reject
H10	between snowy and non-snowy hours	0.862	Do not reject
H11	across wind speed groups	<0.001	Reject
H12	stations	<0.001	Reject

These results support the main EDA interpretation: timing and service structure are the strongest sources of variation in the sample, while weather contributes additional but not uniformly strong signal.

6 Modeling Results and Error Analysis

6.1 Model Comparison

All tuned models improve over the majority-class baseline, which confirms that the selected temporal, operational, and weather features contain predictive signal. XGBoost has the best overall test performance in this experiment: 0.648 accuracy, 0.642 delayed-class F1, and 0.710 ROC-AUC. The decision tree has the highest delayed-class recall, but it gives up more specificity, meaning it predicts delays more aggressively.

Table 3: Test-set model comparison.

Model	Accuracy	F1 delayed	Recall delayed	ROC-AUC
Baseline Majority	0.526	0.000	0.000	0.500
Logistic Regression	0.636	0.623	0.633	0.688
KNN	0.639	0.619	0.618	0.696
Decision Tree	0.635	0.656	0.732	0.683
Random Forest	0.638	0.640	0.679	0.690
XGBoost	0.648	0.642	0.665	0.710

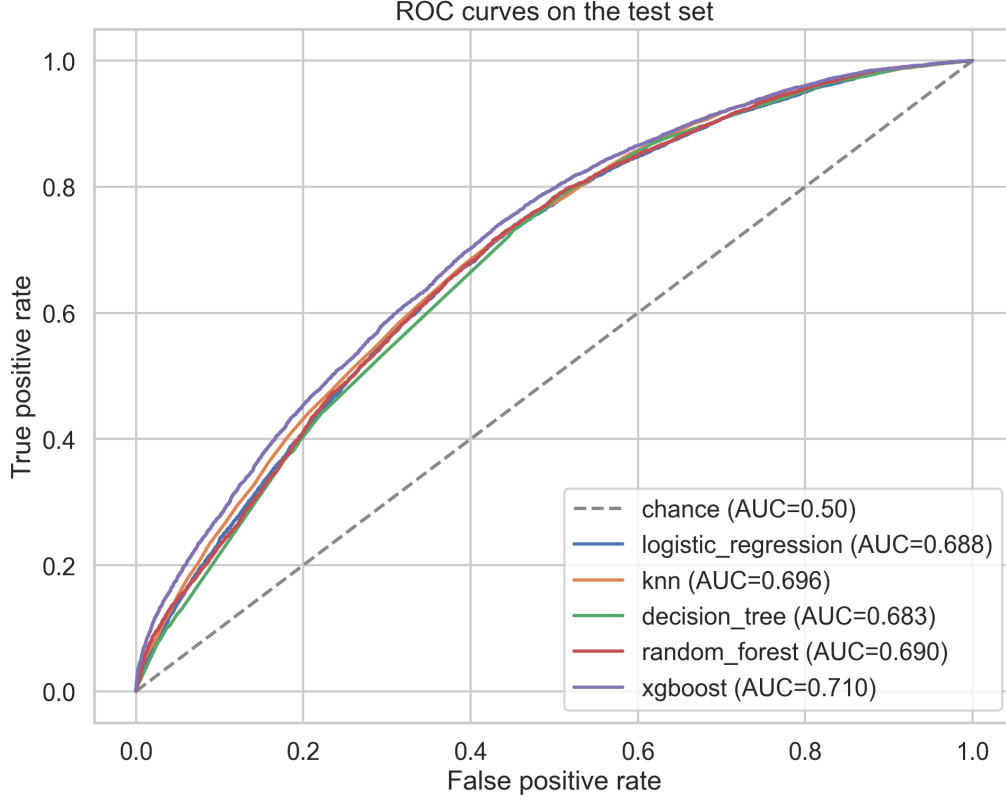


Figure 3: ROC curves for the evaluated models.

6.2 Overall Error Profile

For the selected XGBoost model, the confusion matrix shows 6,037 true negatives, 3,515 false positives, 2,885 false negatives, and 5,727 true positives on our test set with 18,164 rows. This means the model catches about two thirds of actual delayed arrivals, but it still creates many false delay alarms.

Table 4: XGBoost confusion counts on the test set.

	Predicted on-time/early	Predicted delayed
Actual on-time/early	6,037	3,515
Actual delayed	2,885	5,727

6.3 Where the Model Struggles

I grouped test set predictions by scheduled arrival hour and by broad arrival periods. Figure 4 and Table 5 show that the model has different failure modes at different times of day.

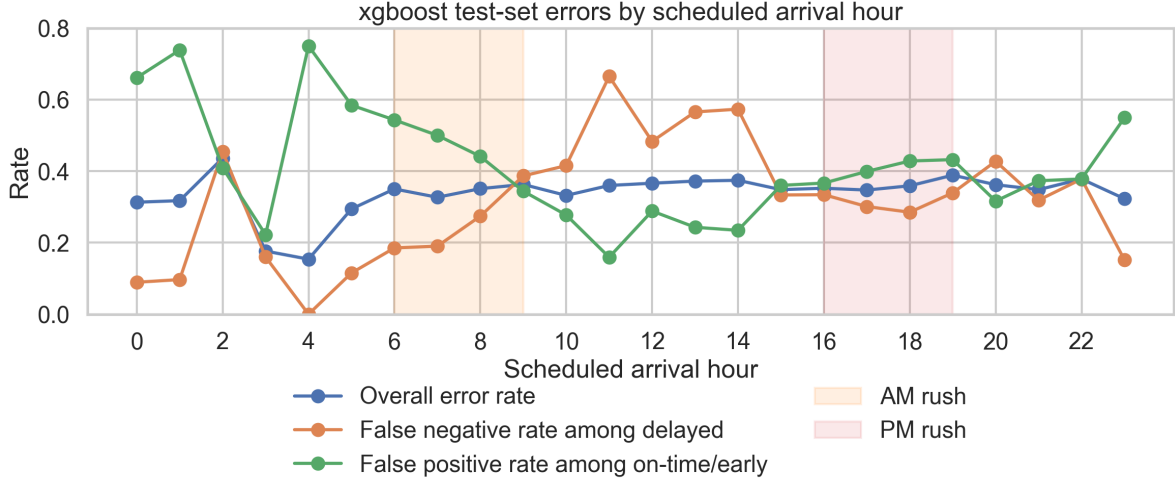


Figure 4: XGBoost error rates by scheduled arrival hour.

Table 5: Error rates by broad scheduled arrival period. arrivals.

Period	Rows	Delay rate	Error rate	FN rate	FP rate
AM rush (06-09)	3,867	52.0%	34.7%	25.1%	45.1%
Midday (10-15)	5,848	40.8%	35.9%	50.2%	26.0%
PM rush (16-19)	3,996	48.0%	36.2%	31.4%	40.6%
Evening (20-23)	3,241	47.2%	35.4%	32.1%	38.3%
Night/early (00-05)	1,212	62.9%	30.4%	11.4%	62.4%

The highest hourly error rate occurs at 19:00, where 38.9% of test rows are misclassified. This hour belongs to the PM rush window. AM rush arrivals have a 34.7% overall error rate, but the main issue is false positives: 45.1% of on-time or early AM-rush arrivals are predicted as delayed. PM rush behaves similarly, with a 40.6% false-positive rate among on-time or early arrivals.

Midday has a different failure mode. Its overall error rate is 35.9%, but the false-negative rate among actual delays is 50.2%. In other words, during midday the model is more likely to miss real delays. The hour specific table makes this visible: at 11:00, the model misses 66.5% of actual delays, and at 13:00 and 14:00 it misses more than half.

Table 6: Hardest scheduled arrival hours, limited to hours with at least 200 test rows.

Hour	Rows	Delay rate	Error rate	FN rate	FP rate
19:00	983	46.8%	38.9%	33.9%	43.2%
22:00	709	48.5%	37.8%	37.8%	37.8%
14:00	959	41.3%	37.4%	57.3%	23.4%
13:00	954	40.0%	37.2%	56.5%	24.3%
12:00	964	39.9%	36.6%	48.3%	28.8%
09:00	951	44.1%	36.3%	38.7%	34.4%
20:00	1,022	41.2%	36.2%	42.8%	31.6%
11:00	978	39.7%	36.0%	66.5%	15.9%

Other slices show that the model can also struggle with particular service contexts. Lausanne-Chauderon has the highest station-level error among stations, mostly because the model overpredicts delay for arrivals that are actually on time or early. RE trains show a similar heavy false-positive pattern.

Table 7: Highest-error groups across selected slices.

Slice	Hardest group	Rows	Delay rate	Error rate	FN rate	FP rate
Station	Lausanne-Chauderon	2,184	56.4%	46.6%	18.8%	82.4%
Train type	RE	1,798	61.3%	38.3%	4.9%	91.2%
Rain condition	Dry	14,277	45.6%	35.6%	37.8%	33.7%
Weekend status	Weekday	13,569	49.5%	35.5%	28.6%	42.3%

Overall, the error analysis suggests that the current feature set captures broad delay risk but not enough train-specific context. It knows that some hours, stations, and train types are risky, but it does not know whether a specific train is already carrying upstream delay, whether a connection conflict is present, or whether disruption information changed shortly before arrival.

7 Limitations and Future Work

The main limitation is that the dataset is arrival-level but is not aware of the trip history. The model sees the scheduled hour, line, train type, station, and weather, but it does not see the train’s

earlier stops or whether it was already delayed upstream. This is likely why performance remains moderate even for the best model. A future version should reconstruct train trajectories and add upstream delay features.

A second limitation is the seasonal sampling design. The project uses four 30-day windows instead of a full year. This gives seasonal coverage, but it may miss unusual disruptions, holiday effects, construction periods, and extreme weather days outside those windows. A full-year sample would make the seasonal and weather findings more stable.

The weather merge is hourly, so it cannot capture short bursts of precipitation, platform-level conditions, or localized disruptions. Adding more granular weather, event, maintenance, and service-alert data could improve both explanation and prediction.

There are also modeling limitations. The final prediction target treats every positive delay as delayed, even if the delay is only one minute. A future extension could compare multiple thresholds, such as delays greater than 3, 5, or 10 minutes. This would better separate operationally meaningful delays from minor timetable noise.

Finally, the analysis is geographically focused on Lausanne. That makes the project coherent, but the conclusions should not automatically be generalized to all Swiss stations. A broader study could compare Lausanne with other stations and evaluate whether the same features matter across different rail network contexts.